

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\sarah\OneDrive\Documents\GitHub\hw5-sarah_nathan_hw5`
Installed Measures          v0.3.3
Installed HiGHS_jll         v1.12.0+0
Installed OpenBLAS32_jll    v0.3.29+0
```

```

Installed GR_jll          v0.73.18+0
Installed PlotUtils       v1.4.4
Installed OpenSSL         v1.6.0
Installed MutableArithmetics v1.6.7
Installed FFMPEG          v0.4.5
Installed Pango_jll       v1.57.0+0
Installed METIS_jll       v5.1.3+0
Installed StaticArraysCore v1.4.4
Installed JSON            v1.3.0
Installed DataStructures  v0.19.3
Installed StatsBase       v0.34.8
Installed FFMPEG_jll      v8.0.0+0
Installed HiGHS           v1.20.1
Installed StableRNGs      v1.0.4
Installed StructUtils     v2.6.0
Installed ForwardDiff     v1.3.0
Installed GR              v0.73.18
Installed JuMP            v1.29.3
Precompiling project...
 8315.9 ms METIS_jll
 8752.8 ms Measures
7049.6 ms OpenBLAS32_jll
4816.6 ms StableRNGs
4789.8 ms StaticArraysCore
4768.7 ms StructUtils
5056.3 ms ChainRulesCore
5218.5 ms DataStructures
1220.6 ms DiffResults
2636.8 ms HiGHS_jll
1619.7 ms StructUtils → StructUtilsTablesExt
3923.8 ms ColorVectorSpace → SpecialFunctionsExt
5145.6 ms OpenSSL
3603.8 ms ChainRulesCore → ChainRulesCoreSparseArraysExt
13485.7 ms MutableArithmetics
3382.8 ms LogExpFunctions → LogExpFunctionsChainRulesCoreExt
1544.9 ms SortingAlgorithms
10090.5 ms ColorSchemes
9791.8 ms Pango_jll
5283.2 ms SpecialFunctions → SpecialFunctionsChainRulesCoreExt
7964.5 ms JSON
13002.9 ms StaticArrays
14511.0 ms FFMPEG_jll
7959.8 ms ForwardDiff

```

6386.4 ms	StatsBase
3940.7 ms	BenchmarkTools
6175.3 ms	StaticArrays → StaticArraysStatisticsExt
6224.5 ms	Adapt → AdaptStaticArraysExt
7293.0 ms	StaticArrays → StaticArraysChainRulesCoreExt
7369.2 ms	ArnoldiMethod
4016.3 ms	ForwardDiff → ForwardDiffStaticArraysExt
11310.2 ms	FFMPEG
16512.0 ms	PlotUtils
8194.3 ms	GeometryTypes
6014.8 ms	Interpolations
20723.7 ms	GeometryBasics
14067.0 ms	GR_jll
9142.0 ms	PlotThemes
30147.2 ms	HTTP
11197.0 ms	Graphs
5286.1 ms	Interpolations → InterpolationsForwardDiffExt
4805.0 ms	NetworkLayout
10359.7 ms	RecipesPipeline
4354.1 ms	NetworkLayout → NetworkLayoutGraphsExt
8507.8 ms	GR
7386.6 ms	GraphRecipes
66066.4 ms	DataFrames
6272.6 ms	Latexify → DataFramesExt
63904.0 ms	MathOptInterface
4658.0 ms	MathOptIIS
16468.9 ms	HiGHS
28860.4 ms	JuMP
89104.4 ms	Plots
7363.3 ms	Plots → GeometryBasicsExt

54 dependencies successfully precompiled in 173 seconds. 198 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
# Data

# City waste generation (mg/d)
cities = 1:3
waste = Dict{1 => 100.0, 2 => 90.0, 3 => 120.0}

# Distances from facilities to cities (km)
dist_city = Dict(
    :LF => Dict{1 => 5.0, 2 => 15.0, 3 => 13.0},
    :MRF => Dict{1 => 30.0, 2 => 25.0, 3 => 45.0},
    :WTE => Dict{1 => 15.0, 2 => 10.0, 3 => 20.0}
)

# Distances between facilities (km)
dist_fac = Dict(
    (:MRF, :LF) => 32.0,
    (:MRF, :WTE) => 15.0,
    (:WTE, :LF) => 18.0
)

# Capacities (Mg/day)
cap = Dict{:LF => 200.0, :MRF => 350.0, :WTE => 210.0}

# Fixed costs ($/day)
```

```

F = Dict(:LF => 2000.0, :MRF => 1500.0, :WTE => 2500.0)

# Tipping costs ($/mg)
tipping = Dict(:LF => 50.0, :MRF => 7.0, :WTE => 60.0)

transport = 1.5      # Transportation costs ($/mg-km)
recycle = 40.0       # Recycling costs ($/mg recycled)

MRF_recycle = 0.4    # MRF recycling rate
ash_NR = 0.16        # Ash fraction of non-recycled waste
ash_recycle = 0.14   # Ash fraction of recycled waste

# Composition table (mass share, ash %, MRF recycle %)
# Component => (mass_share, ash_frac, recycle_frac)
comp = Dict(
  :Food          => (0.15, 0.08, 0.00),
  :PaperCardboard => (0.40, 0.07, 0.55),
  :Plastics       => (0.05, 0.05, 0.15),
  :Textiles       => (0.03, 0.10, 0.10),
  :RubberLeather  => (0.02, 0.15, 0.00),
  :Wood           => (0.05, 0.02, 0.30),
  :Yard           => (0.18, 0.02, 0.40),
  :Glass          => (0.04, 1.00, 0.60),
  :Ferrous        => (0.02, 1.00, 0.75),
  :Aluminum       => (0.02, 1.00, 0.80),
  :OtherMetal     => (0.01, 1.00, 0.50),
  :Misc           => (0.03, 0.70, 0.00)
)

```

Dict{Symbol, Tuple{Float64, Float64, Float64}} with 12 entries:

```

:Plastics      => (0.05, 0.05, 0.15)
:Aluminum      => (0.02, 1.0, 0.8)
:Glass        => (0.04, 1.0, 0.6)
:Textiles      => (0.03, 0.1, 0.1)
:RubberLeather => (0.02, 0.15, 0.0)
:PaperCardboard => (0.4, 0.07, 0.55)
:Yard          => (0.18, 0.02, 0.4)

```

:Ferrous	=> (0.02, 1.0, 0.75)
:OtherMetal	=> (0.01, 1.0, 0.5)
:Misc	=> (0.03, 0.7, 0.0)
:Wood	=> (0.05, 0.02, 0.3)
:Food	=> (0.15, 0.08, 0.0)

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.